DSGE Model Estimation via VAR IRF Matching: The Short and Long of It *

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Abstract

Following De Graeve and Westermark (2025) and Baumeister (2025), I increase the lag length of the VAR in Altig et al. (2005) to re-estimate their DSGE. Moving from $p = 4$ to $p = 12$ lags, I find that both the VAR and DSGE impulse responses, and the implied structural parameters, change substantially. The Calvo price parameters that distinguish the observationally equivalent firm-specific and homogeneous capital versions of the model converge to one another, from 0.34 against 0.82 at $p = 4$ to 0.89 against 0.90 at $p = 12$. The implied duration of price contracts in the firm-specific model rises from 1.5 to 9.0 quarters, equal to that of the homogeneous capital model, re-opening the micro–macro discrepancy in price stickiness that firm-specific capital was introduced to resolve.

Keywords: Truncation Bias, Long-Lag VARs, DSGE Estimation, IRF Minimum Distance , Nominal Rigidities
JEL Codes: C32, C52, E32

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1 Introduction

1.1 Motivation

An important goal of monetary business cycle research is to explain the empirical evidence on the impact of economic shocks on macroeconomic variables. Because the systematic component of policy is endogenous, identifying the causal effect of monetary policy requires isolating the non-systematic component, the monetary policy shock, even though it is small and infrequent (Ravenna, 2007; Aruoba and Drechsel, 2024).

A large class of estimated DSGE models is taken to the data through this channel. Impulse Response Function Matching Estimation (IRFME) is a limited-information strategy that chooses structural parameters to minimise the distance between the impulse responses implied by the model and those estimated from an identified VAR (Christiano et al., 2005; Altig et al., 2011; Rossi et al., 2012). The strategy is attractive precisely because it is partial: the researcher takes a stand on the identification of a subset of shocks, and remains agnostic about the remaining dimensions of the data.

That attractiveness rests entirely on one maintained assumption: that the VAR delivers *consistent* estimates of the moments being matched. If the reduced-form VAR adequately captures the data generating process, impulse responses are optimal at all horizons; if the VAR is misspecified, the specification error propagates to the structural statistics and compounds at each horizon (Ramey, 2016; Canova and Fosso, 2025) Ravenna (2007) "The bias introduced by truncation can lead to bias in the identification of the structural shock." Altig et al. (2011) say as much themselves: "the quality of our estimation strategy depends on the ability of identified VARs to generate reliable estimates of the dynamic response of economic variables to shocks".[1]

Since a VAR(∞) cannot be estimated on a finite sample, researchers truncate. The resulting VAR(p) is misspecified, and increasingly so the smaller is p (De Graeve and Westermark, 2025; Barnichon et al., 2026; Ravenna, 2007). Ravenna (2007) "the problem facing economistis is whether the number of lags p to be included is reasonable given the lenght of economics time series over which VARs are estimated. When estimating a VAR consistent with DSGE business cycle models it is standard to assume that including few lags is sufficient to provide a reasonable approximation to the exact infinite order VAR representation! De Graeve and Westermark (2025) show that the conventional objection to longer lags, that the additional parametrisation makes confidence intervals prohibitively wide, is incomplete, because reducing misspecification reduces not only bias but also variance. Their conclusion is addressed explicitly to applied researchers: "it is up to the researcher to show that her structural results are robust to significant increases in the lag length, far beyond the lag lengths typically considered in current practice".

[1] Altig et al. (2011, footnote 10, p. 3).

1.2 Contribution and Results

This paper asks what lag truncation does to *structural* estimates, not merely to impulse responses. I replicate Altig et al. (2011) — henceforth ACEL — and re-estimate their DSGE by IRF matching against VARs of increasing lag order.

The contribution is twofold. First, relative to De Graeve and Westermark (2025), I move the question from the reduced form to the structural parameters. Second, and more specifically, I do so in a setting their design deliberately excludes. De Graeve and Westermark (2025) state that "all our experiments are based on two-shock models and two-variable VARs", and that where a candidate model does not fit that framework — Smets and Wouters — "we consider the model with only monetary policy and preference shocks".[2] Their data generating processes are square and fundamental by construction, so that truncation bias is isolated from singularity and invertibility. ACEL's model is not: it confronts three structural shocks with a ten-variable VAR. [**TBD:** Verificare che nessun altro abbia gia' fatto questo esercizio su un modello stocasticamente singolare. Se confermato, e' il claim centrale.]

My main results are the following. *First*, the impulse responses to a monetary policy shock retain their sign and broad shape across lag orders, but the humps peak substantially later as p increases, and the point estimates remain inside the VAR(4) confidence bands. *Second*, the responses to the two technology shocks — identified with long-run rather than short-run restrictions — do not: several fall outside the baseline confidence set. This asymmetry is what Ravenna (2007) predicts, since long-run restrictions require the sum of the entire lag polynomial and truncation error accumulates in exactly that object. *Third*, the structural estimates move. The Phillips curve slope falls from 0.0402 to 0.0129 and becomes precisely estimated only at $p = 12$; the cost of varying capital utilisation collapses from 1.99 to 0.004; the investment adjustment cost roughly doubles. *Fourth*, and as a consequence, the implied Calvo price parameters of the firm-specific and homogeneous capital models converge, from 0.34 against 0.82 to 0.89 against 0.90. The implied duration of price contracts in the firm-specific model rises from 1.5 to 9.0 quarters.

The last result overturns ACEL's central conclusion. Their claim is that only the firm-specific capital model reconciles macro evidence on inflation inertia with micro evidence on the frequency of price adjustment. Under a richer lag structure that reconciliation disappears. The channel is transparent and internal to their own argument: ACEL note that "to the extent that capital utilization rates can easily be varied, the assumption of firm-specific capital loses its ability to rationalize low values of ξ_p".[3] Long-lag VARs push the estimate of σ_a precisely into that region.

[2] De Graeve and Westermark (2025, p. 10). They also note that "invertibility is never a problem; all the models we consider are fundamental".

[3] Altig et al. (2011, p. 227).

1.3 Related Literature

Long-lag VARs.
Truncation bias and the DSGE–VAR mapping.
Impulse response matching and DSGE estimation.

1.4 Outline

Section

2 The Theoretical Model

2.1 Inflation inertia and the micro–macro conflict

Altig et al. (2011) start from a tension in the New Keynesian literature. To account for inertia in inflation, standard models require firms to re-optimise prices infrequently — every six quarters, and in versions without indexation to past inflation even every two years. This sits uneasily with the microeconometric evidence, which finds price changes considerably more frequent. Citare Bils and Klenow (2004), Golosov and Lucas (2007), Klenow and Kryvtsov (2008), che sono i riferimenti di ACEL. Valutare se aggiungere evidenza micro piu' recente. Da aggiungere a biblio.bib.

Altig et al. (2005)[4] propose a resolution. In their model inflation is inertial even though firms re-optimise on average every 1.5 quarters. Inertia arises not because firms change prices rarely, but because when they do they change them by a small amount: each firm's short-run marginal cost is increasing in its own output, which dampens the optimal price adjustment. The underlying assumption that drives this result is that capital is predetermined within the period and firm-specific, so that factors of production cannot be instantly and costlessly reallocated across firms.

2.2 Homogeneous and firm-specific capital

Altig et al. (2005) conduct the analysis using two versions of the Christiano et al. (2005) model: one in which capital is homogeneous, as in standard formulations, and the other in which it is firm-specific.

Once log-linearised, the only difference between the two sets of equilibrium conditions lies in the equation relating inflation to marginal cost, the New Keynesian Phillisp Curve,

$$\Delta\hat{\pi}_t = \mathbb{E}\big[\beta\Delta\hat{\pi}_{t+1} + \gamma\hat{s}_t \mid \Omega_t\big], \tag{1}$$

where $\hat{s}_t$ is economy-wide average real marginal cost in units of the final good, and the slope γ maps into the structural parameters as

$$\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\,\chi. \tag{2}$$

The information set Ω_t contains the current realisation of the technology shocks but not that of the monetary policy innovation. The two models are parametrised in terms of γ rather than ξ_p.

Given the remaining parameters they are then observationally equivalent with respect

[4]The replicaiton code made available from the authors is NBER working paper from 2005. Hence, I replicate and refere to those results.

to aggregate prices and quantities, so that no stand need be taken on which version is being estimated. What distinguishes them is the value of ξ_p given γ implies.

The real rigidity parameter χ measures the strategic complementarity in price setting of Ball and Romer (1990). It takes values in $(0, 1]$, and the closer it lies to zero the stronger the attenuation. That is, a firm that is able to reset its price moves it by less, for a given movement in economy-wide marginal cost. In the homogeneous capital model $\chi = 1$ and the attenuation is absent. In the firm-specific model χ is a non-linear function of the structural parameters.

In particular χ depends on some calibrated parameters, $[a, \beta, \delta, \lambda_f]$ which are respectively, capital component in the Cobb-Douglas production function, time preference parameter, capital depreciation rate, and the average price markup. And two estimated parameters, $[\sigma_a; S'']$, which are the Capacity utilisation cost and Investment adjustment cost.

The Altig et al. (2011) logic suggest that, for a fixed value of ξ_p, the larger σ_a, the lower is γ, in line with the vast set of literature that suggests that the Phillips curve slope is low. But other things equal, a lower ξ_p implies a higher γ. This determine that for a given value of γ , ξ_p is a decreasing function of σ_a.
In this sense, a flat Phillips curve is a problem in the homogenoues capital model. A low value of γ implies that, given that there $\chi = 1$, the Clavo price stickiness parameter ξ_p need to be very high, implying irrealistic implied time duration of prices, that is $D = \frac{1}{(1-\xi_p)}$ which are in stark conflict with micro evidence.

The flexibility of capital utilisation is governed by σ_a, and $1/\sigma_a$ is the elasticity of utilisation with respect to the rental rate of capital. Its role is easiest to see by following a firm that has just cut its price and must now meet higher demand. FORSE E' MEGLIO GIRARE IL DISCORSO E PORLO COME SE L'IMPRESA VORREBBE AUMENTARE IL PREZZO Extra labour can be hired at the going wage, but extra capital services can only come from running the capital the firm already holds more intensively, and σ_a is what that costs. If σ_a is high that margin is expensive, so most of the additional output has to be produced by adding labour to a quantity of capital services that barely moves. The firm's own marginal cost then rises sharply with its own output, and a firm that anticipates this cuts its price by less in the first place. That restraint is what χ records, and it is why the firm-specific model needs a high σ_a: only then is χ far enough below one for a low γ to be consistent with a low ξ_p.

It is worth being precise here, because σ_a is a single parameter, estimated once, and does not take different values across the two models. What differs is whether it reaches χ at all. Under homogeneous capital the firm buys capital services on a market at the going rental rate, so its own marginal cost does not depend on its own output and $\chi = 1$

whatever σ_a happens to be. Under firm-specific capital the firm is tied to the capital it already owns, and σ_a determines how much that constraint is actually binding.

The limiting case makes the point. If σ_a is close to zero, utilisation is almost free to vary, so a firm tied to its own capital can still draw from it whatever volume of capital services it needs, at negligible cost. Owning the capital rather than renting it then makes no practical difference: the firm's own marginal cost is as flat as it would be with a market for capital services, χ returns to one, and the firm-specific model collapses onto the homogeneous one. Firm specificity is an assumption about ownership, but its economic content comes entirely from the cost of varying utilisation, and without that cost the constraint exists in name only.

This is where the mechanism is fragile, and Altig et al. (2011) say themselves so: "to the extent that capital utilization rates can easily be varied, the assumption of firm-specific capital loses its ability to rationalize low values of ξ_p".[5] The reconciliation of micro and macro evidence rests, in the end, lies on a single estimated parameter.

2.3 DSGE estimation via IRF matching

Theoretical models are usefull to economists and policymakers only if they are able to reproduce important feauters of the observed data. Which is a property that critically depends on the values that model's parameters have attached, and also on how the researcher decide to estimate them Theodoridis (2011).

Folowing Rotemberg and Woodford (1997); Christiano et al. (2005); Iacoviello (2005); Boivin and Giannoni (2006), Altig et al. (2005) use a limited information approach that minimizes the distance between sample and DSGE model generated impulse responses, this is called Impulse Response Function Matching Estimator (IRFME) Rossi et al. (2012). This procedure select and estimate the model's structural parameters that produce the DSGE IRFs, which minimises the distance with IRFs created from the SVAR.

Using an IRFME implies that the researcher is focusing to match on some limited aspects of the DSGE Giacomini (2013); it is choosing an identification strategy for the reduced form VAR in a way that is consistent with the key determinants of the DSGE. The VAR and theoretical IRFs are identified by restrictions implied by the DSGE. Which means that the DSGE admits a structural VAR representation, in this way, sample moments of the VAR are informative for the theoretical model's structural parameters Rossi et al. (2012).

Theodoridis (2011) points on Canova (1994) and (2005) tought that; many DSGE economistis prefer to use partial information estimators rather than full information ones, like maximum likelihood, because they prefere estimates that minimise the economic dis-

[5] Altig et al. (2011, p. 227).

tance rather than statistical fit criteria. Because structual IRF matching estimates deliver selected DSGE moments-statistic as close as possible to those observed in the data. The economist can assess how well the estimated model replicates these targets allowing the researcher to identify possible modelling weaknesses and to infer usefull economic conclusions.

The authors use a four lag medium scale VAR with 10 quarterly series and 3 shocks: Monetary Policy Shock identified with short run Cholesky restriction, and Embodied and Neutral Technology shocks with long run identification à la Shapiro and Watson (1988). As well summarised by Rossi et al. (2012), the theoretical proposed by Altig et al. (2005) impose that the tech shocks are the only one allowed to affect productivity in the long run. In particular, only Capital Embodied shock can affects the price of investment goods. Finally, Monetary Policy shock affects quantities and prices only within a quarter. The VAR is identified such its IRFs respects the DSGE restrictions and estimate the structural parameters.

2.4 SVAR and IRF minimum distance

The SVAR can be recovered by imposing structure on the reduced form,

$$B_0^{-1}B_0Y_t = B_0^{-1}B_1Y_{t-1} + \cdots + B_0^{-1}B_pY_{t-p} + B_0^{-1}w_t, \tag{3}$$

which matches the RF-VAR term by term once $A_i = B_0^{-1}B_i$ for each lag i and

$$u_t = B_0^{-1}w_t. \tag{4}$$

Equation (4) is the bridge between the two representations: the reduced-form residuals are a linear combination of the structural shocks. Taking variances on both sides,

$$\Sigma_u = B_0^{-1}\Sigma_w\left(B_0^{-1}\right)'. \tag{5}$$

Σ_w is the variance-covariance matrix of the structural shocks w_t. These are assumed mutually orthogonal and normalised to unit variance, so $\Sigma_w = I_K$ by construction. This is a normalisation, not an assumption about the economy: what is estimated is not the scale of each shock, but how a given shock propagates through the system, which is entirely encoded in B_0^{-1}.

Y_t is the ten-dimensional vector of quarterly series used by Altig et al. (2005, 2011),

$$Y_t = \begin{pmatrix} \Delta \ln(\text{relative price of investment}_t) \\ \Delta \ln(\text{GDP}_t/\text{Hours}_t) \\ \Delta \ln(\text{GDP deflator}_t) \\ \text{Capacity utilisation}_t \\ \ln(\text{Hours}_t) \\ \ln(\text{GDP}_t/\text{Hours}_t) - \ln(W_t/P_t) \\ \ln(C_t/\text{GDP}_t) \\ \ln(I_t/\text{GDP}_t) \\ \text{Federal funds rate}_t \\ \ln(\text{GDP deflator}_t) + \ln(\text{GDP}_t) - \ln(\text{MZM}_t) \end{pmatrix}. \tag{6}$$

All the variables are quarterly series from the 1959Q2 - 2001Q4 period, more details on data are described in Appendix A.

Identification combines a short-run and a long-run restriction. The monetary policy shock is identified recursively, following Christiano et al. (2005): the interest rate reacts to lagged variables and to a subset of fast-moving contemporaneous variables, but the shock is assumed orthogonal to that information set, which amounts to zero restrictions on B_0. The two technology shocks are identified through long-run restrictions in the tradition of Shapiro and Watson (1988): both neutral and capital-embodied technology shocks are the only shocks allowed to affect the long-run level of labour productivity, and the capital-embodied shock alone is allowed to affect the long-run relative price of investment. The full derivation is in Appendix B.

These restrictions identify three shocks out of ten. Altig et al. (2005) note that the remaining six are left unlabelled. This SVAR identifies selected aspects that are consistent with the DSGE's shocks. IRFME matches the IRFs of the three identified shocks, not on those not identified.

2.5 Estimating the Parameters of the Equilibrium Model

Using Altig et al. (2005, 2011) notation, let ζ denote the strucutal model parameters that are aimed to be estimated, and $\Psi(\zeta)$ be the mapping from the structural parameters ζ to the model IRFs. The SVAR IRFs estimates are denoted as $\hat{\Psi}$.

The IRFME then can be written as:

$$\hat{\zeta} = \arg\min_{\zeta} (\hat{\Psi} - \Psi(\zeta))' V^{-1} (\hat{\Psi} - \Psi(\zeta)) \tag{7}$$

The IRF matching is done horizon by horizon: the estimator compares the VAR point estimate of each impulse response at each of the first 20 horizons, excluding those set

to zero by identification, against the model-implied counterpart. Here, V is a diagonal matrix composed by the sample variances along the diagonal $\hat{\Psi}$. Those variances, compose the condifence intervals around the point estimates of the SVAR IRFs. With this choice of V, the estimated model's structural parameters $\hat{\zeta}$ are the value of ζ that ensures that the $\hat{\Psi}$ DSGE's IRFs lies as much as possible inside the VAR IRFs' confidence intervals. Standard errors of the DSGE's structural parameters $\hat{\zeta}$ are computed using the Deltha method, more detail in the Appendix C

The minimum distance estimator relies on the ability of the VAR to estimate the IRFs. *"if the VAR is misspecified, the specification errors will be computed at each horizon"* point the attention on potential issues in general VAR practice Ramey (2016) p. 83. This is clear to the author which claim *"The quality of our strategy (IRF matching) depends on the ability of identified VARs to generate reliable estimates of the dynamic response of economic variables to shocks"* Altig et al. (2011) p. 3, fn. 10.

3 Truncation Bias and the Case for Long-Lag VARs

Many DSGE's log linearised solution can be mapped in State Space representation, and then, under some assumption can be re-written as VAR(∞) Fernández-Villaverde et al. (2007). Standard lag choice by SVAR practitioner is to use forecasting information criteria (IC) such as AIC or BIC, but the most common choice is use conventions, that is, 4 lags for quarterly data and 12 for monthly data. If a DSGE is estimated using an IRF matching tecnique, using a short lag VAR it might be problematic as the DSGE is estimated by a biased by truncation VAR. VARs are typically small systems with few lags often interpreted as models of the entire economy, IRFs are highly sensitive to variable and lag choice Barnichon et al. (2026); De Graeve and Westermark (2025).

Truncation Bias it might be a potential problem for structual inference.

3.1 Illustration of the problem: Truncation Bias

Following De Graeve and Westermark (2025), consider a data generating process in which y_t depends on two exogenous variables,

$$y_t = x_{1t}\beta_1 + x_{2t}\beta_2 + \epsilon_t. \tag{8}$$

Estimating the relationship while omitting x_{2t},

$$y_t = x_{1t}b_1 + e_t, \tag{9}$$

delivers a biased point estimate unless the regressors are orthogonal, $\mathbb{E}(b_1) \neq \beta_1$, and an upward-biased residual variance, $s^2 > \sigma^2$. Both distortions matter here, and the second is

the one usually forgotten.

The same logic carries over to a VAR. Let

$$Y_t = B_1 Y_{t-1} + \cdots + B_p Y_{t-p} + u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma_u, \tag{10}$$

with

$$B(L) = B_1 L + \cdots + B_p L^p \tag{11}$$

where $p \ll p^*$, where p^* denotes the true lag order.

Truncation biases both the autoregressive coefficients $B(L)$ and the reduced-form covariance matrix Σ_u. Since impulse responses are a function of both, they inherit the bias.

$$\text{IRF} = f\big(B(L), \Sigma_u\big), \tag{12}$$

The misspecification concerns of short lag VARs which potentially suffer from truncation bias naturally links with the IRF matching estimator. Its role is to minimise the distance of the sample IRFs with the theoretical ones, in order to estimate the model's strucural parameters. Sample IRFs are the objective that the estimator wants to mimic, but if they are misspecified, the estimator is targeting the strucutal parameters to an object that is biased, consequently obtaining potentially distorted estimates of the strucutal parameters.

In light of recent research branch Li et al. (2024); Olea et al. (2025), they dissmiss the use of VARs in of Local Projections (LP), where almost every argument against VARs lies on being VARs lag truncated. While Long-Lag VARs have excellent theoretical properties Plagborg-Møller and Wolf (2021), are considered unworkable in practice due high variance concerns. Which is what typically keeps practitioners away from using long lags is the standard intuition that unceirtany becomes perveasive. In the sense conentional wisdom thiks that, as the lags increases the number of parameters rapidly escalate, therby reducing the degrees of freedom and making confidence interval width prohibitevely large.

De Graeve and Westermark (2025) explore the trivial solution to short lag VARs, that is, dramatic increase in the lag lenght.

3.2 Long-Lag VARs

They're central point is that standard intuition is just a portion of the story. *"in face of misspecification due to lag truncation, increasing the lag length can actually reduce unceirtanty"* De Graeve and Westermark (2025) p. 2.

As the dynamic structure increases, lag truncation decrease, consequently misspecification reduces. Misspecification reduction has a double effect, the first being the well-known bias reduction, while the second is the variance reduction that works against the impreci-

sion coming from the increased parametrisation. This trade off it applies to all truncate VARs, no matter what the identification strategy is.

To reveal the Long-Lag VARs properties De Graeve and Westermark (2025) generate data of realistic samples size from three DSGE with a MonteCarlo, to estimate VARs of different lag lenght. The three DSGE are Chari et al. (2008) hencefore (CKM), Christiano et al. (2007) (CEV) and Smets and Wouters (2007) (SM). All these models as are well knwon, CKM is the most known critique on truncation bias effects on IFRs identified with long-run restrictions. Whereas CEV defence the use shrot run restriction, and SM becasue uses many macro frictions frequently disscussed in macro. They work under some assumptions; that the identification restriction is correct. The DSGE are fundamental in the Fernández-Villaverde et al. (2007) sense, so invertivility is never a problem. Under this assumption these models have an exact VAR($infty$) representation.

With the MonteCarlo they generate 1000 datasets of realistic macro sample size for quarterly data, $T = 200$, and they estimate the bivariate VARs [6] representations of the DSGE choosing different increasing the lag choice When comparing VARs of different lag-lenght, they restrict the sample of effective observations as $T = 170$ ensuting that lag inizialization does not affect sample size.Cosa che io non farò, e difenderò dicendo che tanto le stime non cambiano di molto, e che anzi rimandono consistenti, o che comunque in generale in pratica poi si usano tutti i dati disponibili, non si butta via niente, quindi se poi ai fatti uno deve decidere tra un VAR 4 e un VAR 12 non si mette, nel VAR 4, a buttare via 8 osservazioni per avere lo stesso numero di osservazioni del lag 12, meglio usare tutti i dati. in particolare questo discorso vale ancora di più nel momento in cui si aumenta drasticamente il numero di lag (come negli esercizi di De Graeve and Westermark (2025), e non nel mio) Unceirtanty bands are computed as in (Sims and Zha 1999, Canova 2007, Hulig 2005.), in this sense thy use a the Normal-Inverse Wishart prior, which have good aproximation to frequentist bootstapped intervals. It doesn't different weigth on short vs. long lags as would do other prios like the Minnesota one.

4 Empirical Results

4.1 DSGE structural parameters

[6] For Smets and Wouters (2007), they reduce the number of variables from 7 to 2 to maitanin consistency with the other two exercises. They keep GDP-growth and interest rates.

Table 1: DSGE Structural Parameters Comparison Across Lag Lengths

Parameter	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Structural Estimated Frictions			
Price markup (λ_f)	1.0100	1.0100	1.0100
	(-)	(-)	(-)
Wage stickiness (ξ_w)	0.7221***	0.5617***	0.4838***
	(0.1506)	(0.1142)	(0.0757)
Phillips curve slope (γ)	0.0402	0.0402	0.0129***
	(0.0246)	(0.0598)	(0.0042)
Capacity util. cost (σ_a)	1.9938	0.1175	0.0042
	(2.6066)	(0.6299)	(0.0230)
Habit formation (b)	0.7066***	0.6935***	0.8596***
	(0.0463)	(0.0570)	(0.0203)
Inv. adj. cost (S'')	3.2807*	3.0372*	6.8431***
	(1.8874)	(1.5506)	(2.3306)
Money demand elast. (ϵ)	0.8077***	0.7531***	0.8346***
	(0.2111)	(0.1473)	(0.1944)
Implied Price Stickiness			
Homogeneous capital ($\xi_{p,hom}$)	0.8213	0.8213	0.8958
	(0.0479)	(0.0574)	(0.0139)
Firm-specific capital ($\xi_{p,fs}$)	0.3372	0.6448	0.8884
	(0.1503)	(0.1285)	(0.0220)

Note: Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The Calvo price parameters $\xi_{p,hom}$ and $\xi_{p,fs}$ are not estimated directly: they are recovered numerically from the estimated structural parameters. Their standard errors are therefore obtained by Monte Carlo simulation with 1,000,000 draws from the joint distribution of the estimated structural parameters, solving for the implied Calvo parameter at each draw.

4.2 Implied duration and the collapse of the firm-specific mechanism

Table 2: Implied Average Time between Re-Optimization

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model			
Calvo parameter ($\xi_{p,hom}$)	0.82	0.82	0.90
Implied duration (quarters)	**5.6**	**5.6**	**9.6**
Firm-Specific Capital Model			
Calvo parameter ($\xi_{p,fs}$)	0.34	0.64	0.89
Implied duration (quarters)	**1.5**	**2.8**	**9.0**

- **Baseline VAR ($p = 4$)**: Homogeneous capital implies price changes every **5.6 quarters**.
- **Firm-specific capital**: Drastically reduces this to **1.5 quarters**, matching microeconomic evidence.

Note: Implied duration is calculated as $D = 1/(1 - \xi_p)$.

The Chi parameter

Table 3: Implied Strategic Complementarity Parameter χ Across Lag Lengths

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model (validity check, $\chi \equiv 1$)			
χ_{hom}	0.9989	1.0010	0.9977
	(0.0000)	(0.0000)	(0.0000)
Firm-Specific Capital Model			
χ_{fs}	0.0307	0.2030	0.8671
	(0.0596)	(0.1667)	(0.1698)

Note: χ is recovered by inverting $\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\chi$ at the estimated γ and implied ξ_p. $\chi_{hom} \equiv 1$ by construction (Proposition 1(i) in Altig et al., 2011); reported here only as a check that the numerical pipeline is consistent. Standard errors in parentheses are computed from the same 1,000,000-draw joint Monte Carlo used for the Calvo parameters' standard errors, applying the formula above draw-by-draw. The 90% Monte Carlo percentile interval for χ_{fs} is [0.0067, 0.0879], [0.0516, 0.5101], [0.3704, 0.9270] for $p = 4, 8, 12$ respectively; see the discussion of non-normality of the Calvo/chi Monte Carlo distributions in the text.

5 Conclusion

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A Data

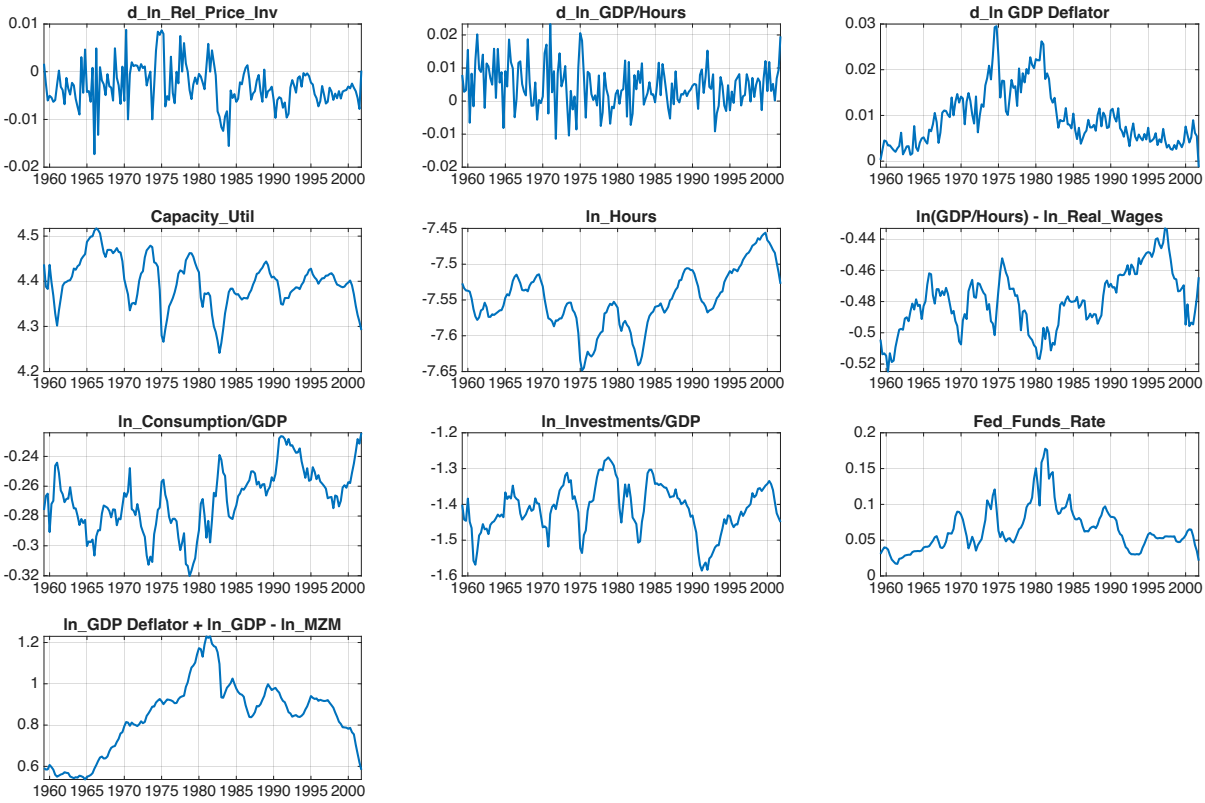


Figure 1: Raw historical series

B Identification strategy

Equation (5) also shows why identification requires restrictions. Σ_u has $K(K+1)/2$ distinct elements, but B_0^{-1} has K^2 free entries, so $K(K-1)/2$ restrictions on B_0 are needed before B_0^{-1} can be recovered from Σ_u. Being numerically sufficient is not enough: the restrictions have to be economically motivated, and it is exactly this requirement that the DSGE is used to satisfy.

C Delta Method

D The ACEL Model: Full Derivation

E Additional Residual Analysis

F Monte Carlo Inference on the Calvo Parameters

G Robustness

H Replication Files and Code